Siemens FTGS Track Circuit
Insulated Rail Joint (IRJ) Inspection

Resistance Measurement, Fault Classification, and Replacement Criteria

Document ID:	MAN-TC-001
Asset Type:	Track Circuit
Language:	EN
Revision:	Rev D | 2025-03-01
Status:	APPROVED FOR USE

1. Purpose and Background

The Insulated Rail Joint (IRJ) provides electrical isolation between adjacent track circuit sections. Degradation of the insulation material allows current to leak between sections, reducing the voltage received by the track circuit receiver. In severe cases this causes false occupancy signals (section reported occupied when clear) or, more critically, the masking of genuine occupancy (train present but not detected). Regular IRJ resistance measurement is therefore a safety-critical maintenance activity.

[Schematic: IRJ cross-section: end posts, fish plates, and insulation material]

Figure: IRJ cross-section: end posts, fish plates, and insulation material

2. IRJ Resistance Specifications

Measure IRJ resistance using a calibrated insulation resistance tester (megohmmeter) at 500 V DC. Take measurements in both dry and wet conditions where possible. Both fish plate joints at a given IRJ location must be measured separately.

Condition	Minimum Resistance	Required Action
Dry conditions - serviceable	>= 1.0 Mohm	No action; re-test at next scheduled visit
Wet conditions - serviceable	>= 0.5 Mohm	No action; monitor trend
At risk - schedule replacement	0.1 - 0.5 Mohm (dry)	Schedule IRJ replacement within 14 days
Critical - urgent replacement	< 0.1 Mohm	Replace within 48 hours - Priority 1

WARNING: Resistance below 0.1 Mohm in any conditions constitutes a critical failure. A false clear condition may exist. Notify the control centre immediately, consider imposing a temporary speed restriction, and arrange replacement within 48 hours.

3. Inspection Procedure

Disconnect the bonding wires at both ends of the IRJ before taking measurements to prevent parallel paths from adjacent bonding corrupting the reading. Reconnect and verify continuity after measurement.

- Step 1: Obtain a line blockage and apply track protection for the affected section.
- Step 2: Visually inspect the IRJ. Look for cracked or missing end posts, deteriorated fish plate insulation sleeves, mechanical damage, or vegetation growing through the joint.
- Step 3: Disconnect rail bonding wires at both sides of the IRJ.
- Step 4: Apply 500 V DC from the megohmmeter between one fish plate and the rail. Allow 60 seconds for the reading to stabilise before recording.
- Step 5: Repeat for the second fish plate.
- Step 6: Reconnect bonding wires and test track circuit operation (occupied/clear cycle).
- Step 7: Record readings, date, ambient conditions, and technician ID in the CMMS.

4. Replacement Criteria and Parts

When resistance is below threshold, schedule or execute IRJ insulation replacement using the standard insulated joint kit. The composite kit includes glass-fibre end posts, insulation sleeves, and hardware and is rated for 760 mm rail.

Spare Part	Part Number	Notes
Insulated joint kit - composite	IRJ-KIT-760-CF	Covers one joint (both fish plates)
End post insulation set (pair)	IRJ-EP-760-SET	Replace if cracked or missing
Fish plate insulation sleeve	IRJ-SLV-760-4PK	4 per kit; replace all when replacing joint

NOTE: Estimated replacement time is 120 minutes per joint by a two-person team. Allow additional time for track circuit commissioning after replacement (see MAN-TC-002 for receiver voltage verification procedure).